

ZAYANDEH RUD: EXTINCTION OR REVIVAL? (AN ANALYSIS OF THE ZAYANDEH RUD RIVER'S RESOURCES, CONSUMPTION, AND NEGATIVE WATER BALANCE)

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ABSTRACT

The Zayandeh Rud River, the most important and largest river in the central Iranian plateau, has faced a negative water balance and drought since the early 2000s due to increased water withdrawals and an imbalance between supply and demand. The table of Zayandeh Rud's water resources and consumption reveals three types of negative balances: an explicit negative balance of 182.3 million cubic meters, a hidden negative balance of 776 million cubic meters, and a conditional negative balance of 257.7 million cubic meters. The areas most severely affected by the river's negative water balance and drought include the river's ecosystem and the Gavkhuni International Wetland, followed by agriculture in the basin. The river's dryness, in addition to causing widespread environmental damage and psychological effects on basin residents, has impacted the livelihoods of millions by halting agricultural activities, reducing tourism, and leading to the closure of small processing and service industries. This article examines the river's condition in recent decades, the imbalance between supply and demand, and its negative water balance.

Keywords: Zayandeh Rud River, Central Iranian Plateau, Negative Water Balance, Water Resources and Consumption.

INTRODUCTION

Iran's water resources are divided into eight regions, 38 watersheds, 147 sub-watersheds, and 618 hydrological units. The Zayandeh Rud watershed, identified as the first watershed in the sixth region of the central plateau, has the code 6-1. Other watersheds in this region include Deq-e Sorkh, Rig-e Zarrin, and Kavir-e Siah Kuh. Located in the central Iranian plateau, this watershed covers a vast area, of which 40% is mountainous and high-altitude, while 60% consists of foothills and plains. The general slope of the watershed decreases from west to east, with the elevation gradually reducing towards the east, reaching the desert plains at its eastern boundary.

The water supply in this watershed comes from precipitation and snowmelt runoff from the eastern Zagros Mountains in the headwaters, as well as from springs and tributaries upstream and downstream of the Zayandeh Rud Dam, which collectively form the Zayandeh Rud River. This river, the most important river in the central Iranian plateau, drains the Zayandeh Rud watershed and flows into Gavkhuni Wetland.

The watershed includes fertile lands and plains suitable for agriculture, irrigated by the Zayandeh Rud River through diversion weirs and modern and traditional irrigation networks. To address water shortages in the central desert region of the watershed, approximately 734 million cubic meters of water are annually transferred to the Zayandeh Rud River through the first and second Kuhrang tunnels, as well as 130 million cubic meters from the Lengan and Khadangestan springs.

Currently, the construction of the third Kuhrang dam and tunnel (aiming to transfer approximately 250 million cubic meters annually), the Behesht Abad water transfer project (also aiming to transfer approximately 250 million cubic meters annually), and the Mandegan water transfer project (aiming to

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transfer approximately 280 million cubic meters annually) is underway by the Isfahan Regional Water Company.

In the national water master plan, the Zayandeh Rud basin is designated as the first basin in Region 6, with the code 6-1. This basin is located between 50°02' to 53°24' East longitude and 31°12' to 33°42' North latitude, covering an area of 41,347 square kilometers. Approximately 90.9% of the watershed lies within Isfahan Province, 2.3% in Yazd Province, 3.4% in Fars Province, and 3.4% in Chaharmahal and Bakhtiari Province. In terms of natural divisions, this basin comprises seven sub-basins and 20 hydrological units. The table below shows the breakdown of the basin into sub-basins and hydrological units.

Table 1. Subdivision of the Zayandeh Rud Basin by Sub-basin and Hydrological Unit (in square kilometers)

Unit Area (Km ²)	Unit Code	Hydrologic Unit Name	Sub-basin Area (Km ²)	Sub-basin Code	Sub-basin Name
692	6111	Daran – Damane	4246	6-1-1	Pelasjan
150	6112	Chehel Khane			
993	6113	Buyin and Miyandasht			
1473	6114	Kohrang – Bardeh			
938	6115	Chadegan			
729	6121	Bon and Saman	4023	6-1-2	Shour Dehaghan
3294	6122	Mobarakeh and Semirum Sofla			
2018	6131	Meymeh	5716	6-1-3	Khoshkerud
1496	6132	Alavijeh, Dehagh			
2202	6133	Morchekhourt			
835	6141-1	Karvan	11948	6-1-4	Murgab
1598	6141-2	Najafabad			
520	6142	Mahyar			
3878	6143	Isfahan, Borkhar			
5117	6144	Kuhpayeh, Segzi, and Margh			
2443	6151	Southern Shahreza	4718	6-1-5	Zar Cheshmeh
2275	6152	Jarghoyeh			
1161	6161	Izadkhast	7080	6-1-6	Rahimi
5919	6162	Dastgerd - Esfandaran			
3616	6171	Gavkhuni	3616	6-1-7	Gavkhuni

Climate: Based on the classifications of the Koppel-Geiger climate system, the basin's climate falls within the arid belt. However, the Zagros and Karkas mountains are sources of significant climatic variations throughout the basin and its surrounding areas. As elevation decreases from northwest to southeast, some climatic elements increase (such as temperature and evaporation) while others decrease (such as precipitation and humidity). Thus, the cold and humid climatic identity gradually gives way to arid and hyper-arid conditions. The table below shows the climate of the Zayandeh Rud (Gavkhuni) basin regions.

Table 2. Climatic Conditions of the Zayandeh Rud Basin

Row	Climate Type	Zayandeh Rud River Basin Regions
1	Wet Cold	Representing Chelgerd and its surrounding areas
2	Semi-humid	Representing Buin-Miyandasht and Fereydunshahr
3	Weakly Semi-arid	Representing Daran and Damaneh
4	Moderately Semi-arid	Representing Chadgan and Kron
5	Severely Semi-arid	Representing Mimeh and Eyzdkhast
6	Arid	Representing the plains of Isfahan, Borkhar, Kuhpayeh-Segzi, Najafabad, South Shahinshahr, and Jarkuyeh
7	Hyper-arid	Representing Gavkhuni and Varzaneh

ZAYANDEH RUD RIVER WATER RESOURCES

(A) Up to 1954 (1333 Solar Hijri)

According to the definition provided by the United States Environmental Protection Agency (USEPA), an intermittent stream is one that flows only during specific periods of the year. In contrast, according to the United States Geological Survey (USGS), a perennial stream is one that flows continuously over time. The Zayandeh Rud River and the Gavkhuni International Wetland are the most important river and wetland in the central Iranian plateau watershed and exhibit a perennial flow regime. However, due to human factors, the river's discharge has become seasonal in some study areas. The traditional water resources of the Zayandeh Rud River (natural inflow) until the early 1950s are shown in the table below:

Table 3: Zayandeh Rud River Water Resources up to the early 1950s (Natural Inflow)

Sources	Description
Chelgerd Branch	The Zayandeh Rud River comprises numerous springs and tributaries, the most important of which are: Dimeh Spring, Namak Spring, Zari Water, Cham Darreh Water, Khorber Water, Naleshgaran Water, Darreh Ghazi Water, Tangeh Gazi Water, and Dolat Abad Water. Approximately 70% of the Zayandeh Rud River's inflow upstream of the reservoir dam is supplied by this branch.
Pelasjan Branch	The Pelasjan River originates south of Khansar. Its northeastern branch originates around Daran and Damaneh, while its northwestern branch originates west of Ozon Bolagh, northeast of Fereydunshahr, and around Khamselu. About 30% of the Zayandeh Rud River's inflow upstream of the reservoir dam is supplied by this branch.
Inter-basin Water Resources	In addition to the constant inflow from the two main tributaries of the Zayandeh Rud, other water sources join the river upstream seasonally, particularly during the equinoxes. This phenomenon is known as inter-basin inflow. The amount of this inflow gradually decreases from upstream to downstream.
Floods and Gullies	Numerous channels, locally known as "La," exist along the Zayandeh Rud River. During rainy seasons, these "Las" carry massive floodwaters into the Zayandeh Rud.
Morghab River	The river originates from the Dalan Kuh mountains and the outlet of the Dotu valley. It includes two major sources: the Morghab River and the Borazgerd Spring. In the Flavarjan county area, it joins the Zayandeh Rud from its left bank.

Sources	Description
Dehaghan Shour River	It originates near the village of Hamgin in Shahreza from a spring of the same name. Its main tributary is the Bodjan River, which flows into the Zayandeh Rud from its right bank downstream of the Sararoud riverside park in Mobarakeh county.
Springs	Along the Zayandeh Rud River, numerous springs contribute water, the most important of which include Hardang Spring, Hasan Abad Spring, Rehnan Spring, Qomeish Spring, Ahmad Spring, and others.
Traditional Drainage Systems	In the floodplain areas of the Zayandeh Rud, before the advent of modern irrigation and water management, numerous locally constructed drainage systems existed, known as key, dastkan, nakesh, harzab, etc. These systems channeled their water resources into the Zayandeh Rud River.
Subsurface flows and seepages	Historically, the Zayandeh Rud River flowed in its thalweg on a low-permeability bed. Furthermore, due to the lack of pumps and pumping technology, as well as the absence of deep and shallow wells, the groundwater table—particularly the subsurface water table—was high. Therefore, due to the creation of a positive hydraulic gradient, the riverbed exhibited subsurface inflows and seepage along its entire length.

The average natural flow of the Zayandeh Rud River, until the early 1950s, is estimated to have been between 1.2 and 2.5 billion cubic meters annually.

(B) From 1954 AD to the Present

Since 1954, with the commissioning of the first Kohrang Tunnel, and since 1970, with the commissioning of the Zayandeh Rud Dam and the storage of unexpected floods and runoff, additional water resources have been added to the Zayandeh Rud River.

Table 4- Zayandeh Rud River Water Resources up to the early 1950s (Natural Inflow)

Row	Source	Nominal Capacity (Mm ³)	Status
1	Tunnel 1	243	Executed
2	Tunnel 2	224	Executed
3	Tunnel 3	120	Partially active and partially completed
4	Cheshmeh Langan Tunnel and Cheshmeh Khadangestan Tunnel	130	Executed
5	Wastewater	200	Partially executed
6	Behesht-Abad Tunnel	250	Not executed
7	Mandegan Project	280	Not executed
8	Water transfer from the Sea of Oman	40	Not executed
9	Disposal of drainage through a modern drainage network (Main drains 1 and 2 of the left Nekoabad network, main drain of the right Nekoabad network, main drain No. 1 of the right Abshar network, and culverts, etc.)	200	This drainage network, from the early 1950s to the early 1970s (approximately two decades), drained the basin's groundwater resources into the Zayandeh Rud River.

The current status of the Zayandeh Rud River's resources is as follows:

Table 5. Current Water Resources of the Zayandeh Rud River

Row	Source	Nominal Capacity (Mm ³)	Average Achieved Yield (Mm ³)	Average Unachieved Yield (Mm ³)
1	Natural inflow	895	700	195
2	Tunnel 1	243	200	43
3	Tunnel 2	224	180	44
4	Tunnel 3	120	50	70
5	Cheshmeh Langan Tunnel and Cheshmeh Khadangestan Tunnel	130	65	65
6	Wastewater	200	75	123
7	Behesht-Abad Tunnel	250	0	250
8	Mandegan Plan	280	0	250
9	Water transfer from the Sea of Oman	400	0	400
10	Water transferred from the drainage network	200	6.3	193.7

Due to the implementation of watershed management projects, the settlement of nomads, and the change in lifestyle of upstream residents from nomadic pastoralism to sedentary lifestyles involving industry, agriculture, and services; as well as the expansion of agriculture, the conversion of pastures and natural areas into farmland upstream, encroachments on the bed and banks of the headwaters, and the drilling of wells in the tributaries, the natural flow of the Zayandeh Rud River has been declining since the early 1930s.

ZAYANDEH RUD WATER USES

(A) Up to 1954 AD (1333 AH), the minimum uses of the Zayandeh Rud River, until the beginning of the 1930s, are as follows:

Table 6. Water Uses of the Zayandeh Rud River until the Early 1950s

Row	Consumer	Status	Water Requirement (Mm ³)
1	Isfahan Agriculture	According to French studies, the equivalent of 113,000 hectares (gross).	850
2	Isfahan Water Supply	Based on the 1954 decree, the equivalent of 30 million cubic meters.	30
3	Isfahan Industry	A large number of watermills that returned the water virtually unchanged.	0
4	Sustainability (Environment)	Given the river's constant flow, the creation of a permanently saturated environment in the riverbed, the absence of shallow, semi-deep, and deep wells, and the balance of surface, subsurface, and groundwater, the need for sustainable water management has been negligible.	20
5	Wetlands (Environment)	It has always had water, and no drying up has been reported to date.	300
Total			1200

Except for the wetland's needs, other uses have remained relatively constant over the years. Any surplus has been directed to the Gavkhuni Wetland. Regarding domestic and industrial water use, the approved

figures in the Zayandeh Rud basin water resources and consumption table (shown below) form the basis for calculating domestic and industrial water needs.

Table 7. Drinking and Industrial Consumption and Subscription Fees

Row	Water Uses	Volume (Mm ³)
1	Drinking Water Rights	30
2	Industrial Water Rights	40.3
3	Withdrawals and Allocations for Chaharmahal and Bakhtiari Province	237
4	Isfahan Drinking Water	404
5	Kashan Drinking Water	49
6	Yazd Drinking Water	98
7	Isfahan Industrial Water	159.7
8	Entitlements of Drinking Water and Industrial Water Rights Holders	303
Total		1321

The flow rate of the Zayandeh Rud River, from the tributaries to the Zayandeh Rud Dam and the dam outlet, over the last 24 years has been as follows:

Table 8. Zayandeh Rud Dam Inflow and Outflow Volumes over the Last 52 Water Years

Row	Water Year	Input (Mm ³)	Output (Mm ³)	Row	Water Year	Input (Mm ³)	Output (Mm ³)
1	1971–1972	1379	754	27	1997–1998	1644	1537
2	1972–1973	1203	1055	28	1998–1999	938	1358
3	1973–1974	1053	1073	29	1999–2000	911	1131
4	1974–1975	1141	1142	30	2000–2001	738	562
5	1975–1976	1865	1690	31	2001–2002	1562	1195
6	1976–1977	1021	1384	32	2002–2003	1529	1422
7	1977–1978	1568	1366	33	2003–2004	1492	1630
8	1978–1979	1154	1270	34	2004–2005	1657	1749
9	1979–1980	1794	1544	35	2005–2006	2219	1916
10	1980–1981	1374	1472	36	2006–2007	1872	1929
11	1981–1982	1167	1235	37	2007–2008	840	1270
12	1982–1983	1300	1257	38	2008–2009	897	763
13	1983–1984	972	1167	39	2009–2010	1263	1453
14	1984–1985	999	1159	40	2010–2011	942	824
15	1985–1986	1420	1116	41	2011–2012	887	1124
16	1986–1987	2122	2051	42	2012–2013	1038	1095
17	1987–1988	2190	2194	43	2013–2014	1036	758
18	1988–1989	1591	1910	44	2014–2015	1430	1419
19	1989–1990	1616	1498	45	2015–2016	1130	1352
20	1990–1991	1144	1441	46	2016–2017	1208	1192
21	1991–1992	2079	1486	47	2017–2018	544	579
22	1992–1993	2508	2526	48	2018–2019	1974	1675
23	1993–1994	1482	1818	49	2019–2020	1457	1725
24	1994–1995	1770	1488	50	2020–2021	738	719
25	1995–1996	1554	1627	51	2021–2022	973	1022
26	1996–1997	1202	1582	52	2022–2023	1375	1095

This section includes the calculation of the negative water balance in the Zayandeh Rud basin's water resources and consumption table.

Water Resources and Consumption Table: Following the resolutions of the thirteenth meeting of the Supreme Water Council of Iran, the water resources and consumption table for the Zayandeh Rud basin was approved as follows in the fourth meeting of the Zayandeh Rud Basin Water Management Coordination Council:

Table 9. Zayandeh Rud Basin Resources and Uses

Row	Category	Volume (Mm ³)	Sources	Volume (Mm ³)
1	Environment	176	Natural Regime	845
2	Isfahan's Water Right (Gross Requirement)	419		
3	Isfahan's Water Allocation (Gross Requirement)	655		
4	Drinking Water Right	30	Kohrang	293
5	Industrial Water Right	40.3		
Total		Total	total	1138
1	Water Allocations and Withdrawals of Chaharmahal and Bakhtiari	237	Kuhrang 2	224
2	Water for Isfahan	404	Kuhrang 3	120
3	Water for Kashan	49	Langan Spring and Khadangestan Spring	130
4	Water for Yazd	98	Behesht Abad	250
Total		Total	total	724
1	Isfahan Industry	159.7	Wastewater	200
2	Portion of Drinking Water, Industrial Water, Water Rights Holders, Shareholders, and Joint Rights	303	Groundwater	303
Expenses Collection		2268	Resource Collection	1862

This table shows three types of negative water balance for the Zayandeh Rud River:

(a) Apparent Negative Water Balance of the Zayandeh Rud Basin:

At first glance, the volume of water consumption in the Zayandeh Rud basin exceeds the volume of its resources. Therefore, a negative water balance is evident in the table, referred to as the apparent negative water balance of the basin. The volume of this is calculated in the table below:

Table 10. Zayandeh Rud River's Apparent Negative Water Balance

Row	Category	Volume of Resources and Expenditures Table (Mm ³)	Resources	Volume (Mm ³)
1	Environment	176	Natural Regime	845
2	Isfahan's Water Right (Gross Requirement)	419		
3	Isfahan's Water Allocation (Gross Requirement)	655		
4	Drinking Water Right	30	Kohrang	293
5	Industrial Water Right	40.3		
Total		1320.3		1138
Obvious negative balance		-182.3		

(b) Hidden Negative Water Balance of the Zayandeh-Rud Basin:

A second perspective reveals that the water balance table omits or underestimates certain water uses, resulting in a negative balance known as a hidden negative balance. For example, agricultural water withdrawals in Chaharmahal and Bakhtiari Province are significantly underestimated, and water requirements for the Ben-Borujen water transfer project, as well as unauthorized well and pump extractions, are not included.

Furthermore, the water sources supplying the allocated water rights of Isfahan Province in the Mahyar and Jarghoyeh plains, Borkhar and Rudashtein (Kuhpayeh-Segzi), namely groundwater resources, are not considered. Firstly, these plains have limited groundwater resources. Secondly, these plains are critically overexploited, severely and catastrophically subsiding, and any groundwater extraction, even if it existed, would be technically unsound, unethical, and illegal. Thirdly, the primary objective of developing irrigation networks in these plains is to rehabilitate and recharge the collapsing aquifers, not to extract water from them. The hidden negative water balance of the Zayandeh-Rud basin is calculated in the table below:

Table 11. Hidden Negative Balance of the Zayandeh Rud River

Row	Category	Planned Volume from Zayandeh Rud (Mm ³)	Actual Required Volume (Mm ³)	Negative Volume (Mm ³)
1	Water Rights in Isfahan Province	237	600	-363
2	Water Transfer to Ben and Borujen	0	195	-195
3	Wells and Encroaching Pumps	0	41	-41
4	Environmental Sustainability of Rivers and Wetlands	0	40	-40
5	Water Rights in Isfahan Province	176	313	-137
Total		237	881	-776

(c) Conditional Negative Water Balance of the Zayandeh-Rud Basin:

The Resources and Consumption table shows all current water users drawing from the Zayandeh-Rud River. However, some listed resources have not yet been realized or made available for exploitation. Therefore, a conditional negative water balance exists in the basin until these resources are realized, as calculated in the table below.

Table 12. Conditional Negative Balance of the Zayandeh Rud Riverd) Total negative water balance of the Zayandeh Rud River:

Row	Category	Planned Volume from Zayandeh Rud (Mm ³)	Actual Required Volume (Mm ³)	Negative Volume (Mm ³)
1	Agricultural Uses in Chaharmahal and Bakhtiari Province	237	600	-363
2	Water Rights in Isfahan Province	0	195	-195
3	Water Transfer to Ben and Borujen	0	41	-41
4	Wells and Encroaching Pumps	0	40	-40
5	Environmental Sustainability of Rivers and Wetlands	176	313	-137
Total		237	881	-776

The total negative water balance of the Zayandeh Rud Basin is presented in the table below:

Table 13. Total Negative Balance of the River

Row	Description	Volume (Mm ³)
1	Apparent Negative Balance	-182.3
2	Hidden Negative Balance	-776
3	Conditional Negative Balance	-257.7
Total		-1216

DECIDING ON HOW TO DIVIDE AND DISTRIBUTE THE RIVER'S WASTEWATER RESOURCES:

Following the decline in water resources in the river basin in the mid-1980s and the subsequent occurrence of crises and social tension in this area, several key developments took place:

- In the mid-2000s (approximately 2006–2007), the management of water distribution for the Zayandeh Rud River was entrusted to the Headquarters of Unexpected Events of Isfahan Province, which was later renamed the General Crisis Management Department of Isfahan Province.
- In the late 2000s (approximately 2009–2010), the Isfahan Provincial Water Station was established with the participation of all stakeholders and took on the responsibility for the division and distribution of water.
- In the early 2010s (approximately 2011–2013), with the provincialization of regional water companies and the independence of the Khuzestan-Bakhtiari Regional Water Company, continuous management of the Zayandeh Rud River watershed was formed and assumed responsibility. During this period, the annual program for the distribution and allocation of the river's water, titled "Resource and Expenditure Table," was compiled and communicated annually by the parent company specializing in Iranian Water Resources, a practice that continues to this day.
- In the mid-2010s (mid-1390s Shamsi), a five-party committee was formed to facilitate the preparation of the Resource and Expenditure Table, which was later renamed the 15-member committee. The first table of resources and expenditures was communicated and implemented in 2010–2011 (1389–1390 Shamsi).

In the late 2010s (late-1390s Shamsi), an ordinance for the formation of the Working Group on Adaptation to Water Scarcity in the provinces was conveyed by the chairman of the provincial governor and the secretary of the Regional Water Company, leading to the establishment of this working group. Currently, the water distribution and allocation programs for the Zayandeh River are being drafted and implemented by the Regional Water Company of Isfahan Province, with participation from the Working Group on Adaptation to Water Scarcity in Isfahan Province, after being notified by the specialized parent company of Iran's Water Resources.

Table 14: Environmental Share in the Basin's Resources and Expenditures from Inception to Present

Row	Year	Total Predicted Inflow (Mm ³)	Environmental Share (Mm ³)	Percentage of Environmental Uses to Total Output (%)
1	2010-2011	806	93	11.54
2	2011-2012	1094	83	7.59
3	2012-2013	1071	31.5	2.91
4	2013-2014	717	12	1.67
5	2014-2015	1187	85	7.16
6	2015-2016	1284	86	6.69
7	2016-2017	1163	87	6.70
8	2017-2018	700	8	1.14
9	2018-2019	1104	49	4.43
10	2019-2020	911	68	7.46
11	2020-2021	685	34	4.96
12	2021-2022	992	78	7.86
13	2022-2023	1023	63	6.16
14	2010-2011	1240	80	6.45

In the announced water resource allocation plans, the allocated volume was solely for meeting the environmental needs of the riverbed and channel, as well as for maintaining flow sustainability. It appears that no allocation was made for the wetland's water rights.

The announced water resource allocation plans were implemented exactly as intended by the Isfahan Regional Water Company, and the company did not deviate from the approved plans regarding environmental needs and maintaining flow sustainability.

However, the announced water resource allocation plans did not adhere to the 176 million cubic meters allocated for environmental needs, as approved in the thirteenth meeting of the Supreme Water Council and the fourth meeting of the Zayandeh Rud Basin Coordination Council.

Based on the resolution of the 31st meeting of the Water Rights Strategy Committee, the environmental water right of the basin is equivalent to 31.09% of the total inflow of the Zayandeh Rud River, which was not honored in the announced water resource allocation plans.

Table 15. Environmental Water Allocation Table (Resolution of the 31st Meeting of the Water Rights Strategy Committee)

Environment	River flow stability	Isfahan's needs	Total
15.46	13.29	2.34	<u>31.09</u>

Table 16. Environmental Water Allocation of the Zayandeh Rud River, Based on Different Methods, and Studies

Row	Method	Amount (Mm ³)	Percentage (%)
1	Studies by Isfahan University of Technology and University of Shahrekord	313	----
2	Ministry of Energy's Water Resources Supply and Demand Table	176	13.33
3	Resolution of the 31st Meeting of the Water Rights Strategy Committee	----	31.09

CONCLUSION

The Zayandeh Rud River faces a negative water balance due to allocations, excessive loads, increased extractions—particularly in the upstream and headwaters—unconventional watershed management practices, industrial development, and agricultural expansion, all contributing to an imbalance between supply and demand.

Based on the resolutions of the thirteenth meeting of the Supreme Water Council of Iran and the fourth meeting of the Zayandeh Rud River Basin Management Coordination Council, a table of water resources and consumption for the Zayandeh Rud River has been prepared and issued.

The Zayandeh Rud water balance table shows three types of negative balances: an apparent negative balance of 182.3 million cubic meters, a hidden negative balance of 776 million cubic meters, and a conditional negative balance of 257.7 million cubic meters.

The areas most severely affected by the negative water balance and river drying are the river's ecosystem and the Gavkhuni International Wetland, followed by agriculture in the basin.

The main priorities for the river's rehabilitation include implementing water transfer projects, saving water and eliminating some unnecessary loads, controlling extractions in the headwaters, adjusting watershed dams, and recycling water resources, especially treated wastewater from sewage treatment plants.

ACKNOWLEDGMENT AND APPRECIATION

We express our sincere gratitude to the Mirab Zayandeh Rud Operation Company for providing the necessary facilities and data for conducting this research and preparing the related article.

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